

§2.1 Hydrostatic Condition

- (1) Velocity of flow is Zero.
- (2) Pressure difference is due only to the weight of the fluid.

如果已知重力场下的某种流体, 其压力可通过积分求解。

Contents of this chapter

- (1) 大气和海洋中的压力分布 (pressure distribution)
 - (2) 压力表的设计 (manometer)
 - (3) 施加于浸没平板和曲面上的力
 - (4) 浸没物体上的浮力 (buoyancy)
 - (5) 流体的性质
- } result in Archimedes' principles

§2.2 pressure and pressure gradient

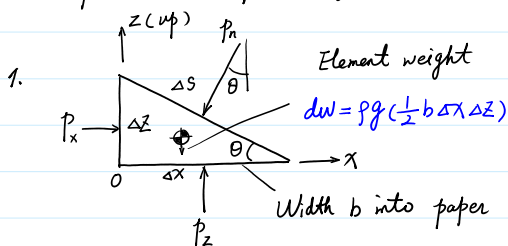


Fig 2.1

- As known:
- (1) fluid at rest, $u=0$.
 - (2) wedge with size of $\Delta x / \Delta y / \Delta z / b$
 - (3) no shear stress.
 - (4) the pressure p_x, p_z and p_n may be different.
 - (5) no acceleration in both x/z directions
 - (6) weight of element must be considered.

$$\text{So: } \sum F_x = 0 = p_x \cdot b \Delta z - p_n b \Delta S \cdot \sin \theta \quad (2.1)$$

$$\sum F_z = 0 = p_z \cdot b \Delta x - p_n b \Delta S \cdot \cos \theta - \frac{1}{2} \rho g b \cdot \Delta x \Delta z$$

According the geometry of the wedge

$$\Delta S \cdot \sin \theta = \Delta z$$

$$\Delta S \cdot \cos \theta = \Delta x \quad (2.2)$$

substitution could be done as:

$$p_x = p_n, \quad p_z = p_n + \frac{1}{2} \rho g \Delta z \quad (2.3)$$

such equations shows two principles
of hydrostatic or shear-free.

- (1) there is no pressure change in the horizontal direction
- (2) there is a vertical change in pressure proportional to Density, gravity and depth change.

在极限情况下，流体三角楔收缩为一个“点”

While $\Delta z \rightarrow 0$, then (2.3) become

$$p_x = p_z = p_n = p$$

Since θ is arbitrary (任意的), so a conclusion will be:

The pressure p at a point in a static fluid is independent of orientation

* what about the pressure at a point in a Moving fluid?

If strain rates (应变率) exist in a moving fluid,

there will be Viscous stresses, both shear and normal in general.

(剪切应力) (法向正应力)

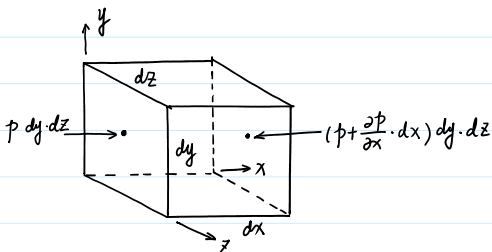
then,

$$p = -\frac{1}{3} (\sigma_{xx} + \sigma_{yy} + \sigma_{zz})$$

(the average of three normal stresses)

"-" sign occurs because a compression stress is considered to be negative whereas p is positive.

2. Pressure Force on a Fluid Element



As known, $p = p(x, y, z, t)$

If pressure vary arbitrarily, the net force in the x direction on the element is given by

$$dF_x = p \cdot dy \cdot dz - (p + \frac{\partial p}{\partial x} \cdot dx) \cdot dy \cdot dz$$

$$= - \frac{\partial p}{\partial x} \cdot dx \cdot dy \cdot dz$$

in similar manner, $dF_y = - \frac{\partial p}{\partial y} \cdot dy \cdot dx \cdot dz$

$$dF_z = - \frac{\partial p}{\partial z} \cdot dz \cdot dx \cdot dy$$

The total net-force vector (向量) is

$$dF_{\text{press}} = (-i \frac{\partial p}{\partial x} - j \frac{\partial p}{\partial y} - k \frac{\partial p}{\partial z}) \cdot dx dy dz \quad (2.8)$$

$\rightarrow$ " - " pressure gradient
Volume

so the net force per unit element volume,

$$f_{\text{press}} = -\nabla p \quad (2.9)$$

such illustrate: It is not the pressure but the pressure gradient causing a net force which must be balanced by gravity or acceleration or some other effect in the fluid.

§ 2.3 Equilibrium of a Fluid Element

① The pressure gradient is a Surface force acts on the sides of the element.

② There may also be a Body force, due to electromagnetic or gravitational potentials (电磁势能) (惯性势能) acting on the entire Mass of the element.

1. If only a gravity force is considered,

$$dF_{\text{grav}} = \rho g \cdot dx dy dz$$

$$\text{or } f_{\text{grav}} = \rho \cdot g$$

通常, 此时还存在由于压力梯度引起的表面力,

如果有的话, 就是粘性应力。

对一种不可压缩流体 (incompressible), 常粘度 (constant viscosity), 粘性力可表达为:

$$f = \mu \left(\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \right) = \mu \nabla^2 V$$

(constant viscosity), 粘性力可表达为:

$$f_{vis} = \mu \left(\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \right) = \mu \nabla^2 \underline{V}$$

velocity

此时作用在流体单元上有三种力:

压力, 重力, 粘性力

或者三种力的合力为零, 保持单元平衡状态 (Equilibrium)

或者合力推动流体单元加速 (move with acceleration)

From Newton's Law, we should have

$$\rho \cdot a = \Sigma f = f_{press} + f_{grav} + f_{visc}$$

$$= -\nabla p + \rho g + \mu \nabla^2 V \quad (2.12)$$

above equation is one of the differential momentum equation
 ↓
 Vector Addition

式中加速度反映了当地的力平衡, 而且加速度并不一定平行于当地的速度矢量, 从而反映了流体了时刻运动的方向。

2. 本章中考题的情况都是已知速度和加速度, 从而求解流体中的压力变化。

Rewrite Eq. (2.12) as

$$\nabla p = \rho(g - a) + \mu \nabla^2 V = B(x, y, z, t) \quad (2.13)$$

↳ vector sum

If V , $a = dv/dt$ are known,

density ρ , viscosity μ are known as well,

we can solve Eq. (2.13) for $p(x, y, z, t)$ by direct integration

$$\begin{cases} \frac{\partial p}{\partial x} = B_x(x, y, z, t) \\ \frac{\partial p}{\partial y} = B_y(x, y, z, t) \\ \frac{\partial p}{\partial z} = B_z(x, y, z, t) \end{cases}$$

如果用方向分量表示就成了
一阶偏微分方程

(注意: $\frac{\partial p}{\partial t}$ 是额外函数)

this extra function comes from a condition of known time variation

$$p_0(t) \Big|_{x_0, y_0, z_0}$$

If it is steady flow, $\frac{\partial p}{\partial t} = \text{Const.}$

$$\text{the } p_0 \Big|_{x_0, y_0, z_0}$$

* Find the pressure distribution from a known velocity distribution is one of the easiest problems in fluid mechanics.

As for Eq. 2.13 we could summarize at least four cases:

(1) Flow at rest or at constant velocity

$$\nabla p = \rho(\mathbf{g} - \cancel{\mathbf{a}}) + \mu \cancel{\nabla^2 \mathbf{V}}$$

hydrostatic

(2) Rigid-body translation and rotation
刚体 (平移) (旋转)

$$\nabla p = \rho(\mathbf{g} - \mathbf{a}) + \mu \nabla^2 \mathbf{V}$$

(3) Irrotational motion ($\nabla \times \mathbf{V} \equiv 0$)
无旋流 (无涡运动)

$$\nabla p = \rho(\mathbf{g} - \mathbf{a}) + \mu \cancel{\nabla^2 \mathbf{V}}$$

Bernoulli's equation

(4) Arbitrary viscous motion

Nothing helpful happens.
no general rules apply.

3. Gage Pressure and Vacuum Pressure

(1) Absolute pressure / total magnitude

(2) Relative pressure to the local ambient atm.

The measured pressure may be either higher or lower than the local atm. each case is given as,

① $p > p_a$ Gage Pressure $p(\text{gage}) = p - p_a$

② $p < p_a$ Vacuum Pressure $p(\text{vacuum}) = p_a - p$

S2.4 Hydrostatic Pressure Distribution

$$\nabla p = \rho \mathbf{g} \quad (2.15)$$

(1) at rest, $\mathbf{a} = 0$, $\nabla^2 \mathbf{V} = 0$

(2) regardless of viscosity.

Eq. 2.15 is a hydrostatic distribution,
is correct for all fluid at rest.

As for the ordinary coordinate system,

z is "up", hence, the local-gravity Vector

for small scale problem is

$$\vec{g} = -g \vec{k}$$

$$g = 9.807 \text{ m/s}^2 \quad (\text{magnitude of local gravity})$$

thus, the components will be,

$$\frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = -\rho g$$

Replaced by $\frac{dp}{dz}$, then,

$$\begin{cases} \frac{dp}{dz} = -\rho g \\ p_2 - p_1 = -\int_1^2 \rho g dz \end{cases} \quad (2.18)$$

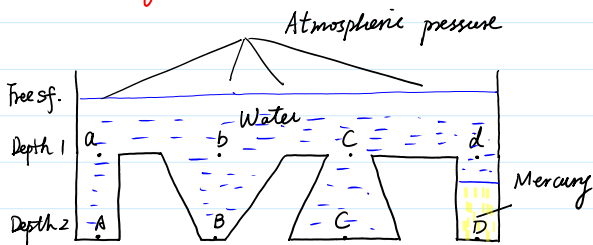
$\xrightarrow{\text{depth}}$

Above equation is the solution of hydrostatic problem.

上述积分方程需要已知密度 ρ 和重力分布的假设, 对于气体和液体通常是不同处理的.

Conclusion of Hydrostatic Condition

- ① Pressure in a continuously distributed uniform static fluid varies only with vertical distance and is Independent of the shape of the container.
- ② The pressure is the SAME at all points on a given horizontal plane in the fluid.
- ③ The pressure increases with depth in the fluid.



2. Effect of Variable Gravity

$$g = g_0 \left(\frac{r_0}{r} \right)^2$$

r_0 — radius of planet

g_0 — the surface value of g .

- ① For the earth, $r_0 \approx 3960$ statute mile

$$1 \text{ statute mile} = 5280 \text{ inch} / 1609.3 \text{ meter}$$

$$r_0 = 3960 \times 1609.3 = 6372.8 \text{ km.}$$

② As for engineering cases,

r : 11 km (deepest ocean) ~ 20 km (超音速飞机飞行高度)

$$\begin{aligned} \left(\frac{6372.8}{6372.8 + 20} \right)^2 &= (0.9969)^2 \\ &= 0.9937 \\ &\approx 99.4\% \end{aligned}$$

hence, we could neglect the variation of g in most problems.

3. Hydrostatic Pressure in Liquids

Recall from Zg. 2.18,

$$p_2 - p_1 = - \int_1^2 \rho g dz$$

1) neglect liquids density variation (incompressible)

$$\text{thus, } p_2 - p_1 = - \rho g (z_2 - z_1)$$

ρ specific weight
比重 N/m^3

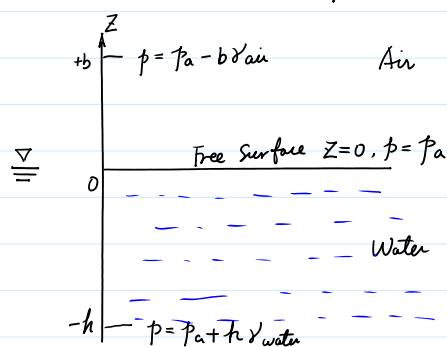


Fig. 2.5

Let $(p_1, z_1) = (p_a, 0)$, then,

for p at any depth z

$$p = p_a - \gamma z \quad (2.21)$$

γ — Average specific weight of lake or ocean.

4. Mercury Barometer

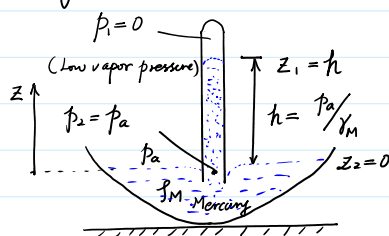


Fig. 2.6

$$\left\{ \begin{array}{l} p_1 = 0, z_1 = h \\ p_2 = p_a, z_2 = 0 \end{array} \right.$$

$$p_a - 0 = -\gamma_M (0 - h)$$

$$h = \frac{p_a}{\gamma_M}$$

at sea-level, $p_a = 101350 \text{ Pa}$

$$\gamma_M = 133100 \text{ N/m}^3$$

$$\text{thus, } h = 0.761 \text{ m} / 761 \text{ mm} \\ = 29.96 \text{ in Hg (U.S.)}$$

Mercury is applied because it is the heaviest common liquid.

While a water barometer would be 34 ft high.

5. Hydrostatic Pressure in Gases

(1) Gases are compressible, $\rho = \rho(p)$

(2) follow the perfect-gas law $p = \rho R T$

According to Eq. (2.18)

$$\frac{dp}{dz} = -\rho g = -\frac{p}{RT} \cdot g$$

$$\int_1^2 \frac{dp}{p} = \ln \frac{p_2}{p_1} = -\frac{g}{R} \int_1^2 \frac{dz}{T} \quad (2.23)$$

Assumption, isothermal atmosphere, $T = T_0$

$$p_2 = p_1 \cdot e^{\left[-\frac{g}{RT_0}(z_2 - z_1)\right]}$$

↳ dimensionless

Actually, the earth's mean atmospheric temperature drops off nearly with z up to an altitude of about

36000 ft. (11000m) $T \approx T_0 - Bz$ (troposphere)

B — lapse rate

对流层

$$T_0 = 518.69^\circ \text{R} = 288.16 \text{ K} = 15^\circ \text{C}$$

$$B = 0.00650 \text{ K/m}$$

$$R = 287 \text{ m}^2/(\text{s}^2 \cdot \text{K})$$

$$\text{thus, } p = p_a \left(1 - \frac{Bz}{T_0}\right)^{g/(RB)} \quad g/RB = 5.26 \text{ (air)}$$

* while in the troposphere, with $z=0$ at sea-level.

If the linear approximation can be evaluated,

$$\left(1 - \frac{Bz}{T_0}\right)^n = 1 - n \frac{Bz}{T_0} + \frac{n(n-1)}{2!} \left(\frac{Bz}{T_0}\right)^2 - \dots$$

$$n = g/RB$$

$$\text{Rearrange, } p = p_a - \gamma_a z \left(1 - \frac{n-1}{2} \cdot \frac{Bz}{T_0} + \dots\right)$$

↳ if this term is small enough

$$\text{which means, } \frac{n-1}{2} \cdot \frac{B}{T_0} z \ll z_0 = 20800 \text{ m}$$

$$z \ll \frac{2T_0}{(n-1)B} = 20800 \text{ m}$$

If z or δz is less than 1000m, then exact error will be within 5%.

* 重力作用下可压缩流体的压强分布

对于气体, 考虑其压缩性, 此时 $\rho = \rho(p)$

不再是常数, 因此若求解可压缩流体中的压强分布, 需要补充 ρ 与其它参数之间的关系。

一般来说:

(1) T_{air} decrease with the height rising

but will be stable at about 11 km high

(到达恒流层以后, 主要是层内臭氧吸收的太阳能热量, 维持该层气温稳定)

(2) 国际宇航协会制定了一套标准的 p , ρ , T 的分布规律, 规定大气层温度分布为:

$$\begin{cases} T = T_0 - Bz, \text{ K} & (z \leq 11000 \text{ m}) \\ T = 216.5 \text{ K} & (z > 11000 \text{ m}) \end{cases}$$

$$T_0 = 288.16 \text{ K} = 15^\circ \text{C} \text{ at sea level}$$

B — 衰减率, 0.0065 K/m

Z.1 Find out Pressure Distribution of International standard atmosphere.

(1) $\rho(p) \Rightarrow$ regarded as idea gas.

follow the rules of $p = \rho R T$

R — Const, $R = 287 \text{ N} \cdot \text{m}/(\text{kg} \cdot \text{K})$

$$\text{Transform, } \rho = \frac{p}{RT} = \frac{p}{R(288.16 - 0.0065z)}$$

According (2.18)

$$\frac{dp}{p} = -\rho g$$

(n-1)B
If z or δz is less than 1000m, then expect error will be within 5%

According (2.18)

$$\frac{dp}{dz} = -\rho g$$

$$\frac{dp}{dz} = -\frac{\rho g}{R(288.16 - 0.0065Z)}$$

$$\int_1^2 \frac{dp}{\rho} = \int_1^2 \frac{-g dz}{R(288.16 - 0.0065Z)}$$

$$\frac{p}{\rho_a} = \left(1 - \frac{0.0065Z}{288.16}\right)^{\frac{g}{0.0065R}}$$

p_a is taken as sea level, $Z=0$,

$$g = 9.807 \text{ m/s}^2$$

$$R = 287 \text{ N}\cdot\text{m}/(\text{kg}\cdot\text{K})$$

Then p distribution in troposphere will be,

$$p = p_a \left(1 - \frac{Z}{44300}\right)^{5.26}, \text{ while } Z \leq 11 \text{ km}$$

When $Z = 11 \text{ km}$,

$$p_1 = p_a \left(1 - \frac{11000}{44300}\right)^{5.26} = 0.223 p_a$$

If p distribution upon troposphere, such as in isothermal layer,

$$\frac{p}{p_1} = \exp \left[\frac{g}{RT} (Z_1 - Z) \right]$$

while, $Z_1 = 11000 \text{ m}$, $p_1 = 0.223 p_a$

$$T = 216.5 \text{ K}$$

Substitute above data into equation, thus

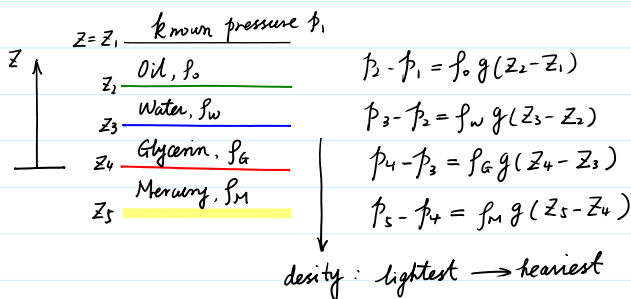
$$p = 0.223 p_a \exp \left(\frac{11000 - Z}{6336} \right), Z > 11 \text{ km}$$

§ 2.5 Application to Manometry

1. From formula 2.20, $p_2 - p_1 = -\gamma(Z_2 - Z_1)$

change in pressure $\leftarrow$ change in elevation of a liquid

Thus, a static column of one or more liquids/gases can be used to measure Pressure Differences between two points. such device is called a **Manometer**.



2. A Memory Device: Up Versus Down

"It is a problem for Engineers to understand

"A pressure increase is downward"

Prof. John Fors of MSU.

$$p_{\text{down}} = p_{\text{up}} + \gamma |\Delta z| \quad (2.32)$$

Just keep adding on pressure increment as moving down through the layered fluid.

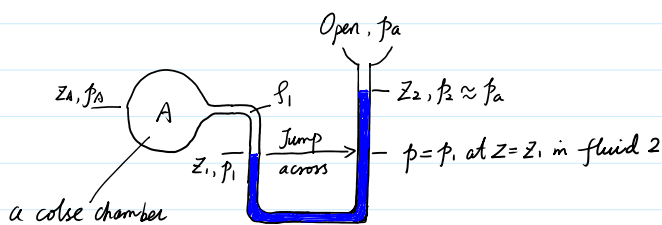


Fig. 2.9

The chamber fluid p_1 is combined with a second fluid p illustration (1) to protect the environment from corrosive chamber fluid.

(2) the open tube will be shorter since a heavier fluid p_2 keeps Z_2 smaller.

According to Eq. 2.32

$$p_A + \gamma_1 |Z_A - Z_1| - \gamma_2 |Z_1 - Z_2| = p_2 \approx p_{\text{atm}}$$

* Pascal's Law needed

Any two points at the same elevation in a continuous mass of the same static fluid will be at same pressure.

§ 2.6 Hydrostatic Forces on Plane Surfaces

1. 在工程实际中有时不仅要知道液体中的压强分布, 还要知道浸没在液体中的物体表面所受压强合力的大小及其作用点。如: 水坝、水闸、港口建筑、储液容器等。

* 作用于平板表面上的流体静力学问题可归纳为: 平板横截面上: 质心与惯性力矩的问题

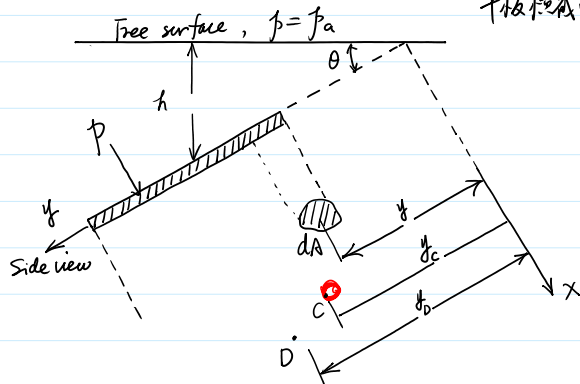


Fig. 2.10

- (1) 一块倾斜平板, 板左侧浸没在静止的液体中, 右侧与大气接触。

平板与水面夹角为 θ 。

- (2) 对平板上任意一微元面积 dA 进行受力分析

液体侧受力为: $p dA = (p_a + \rho g h) dA$

而在平板另一侧有: 空气施加的力 $p_a \cdot dA$

$$\text{则合力 } dF = \rho g h dA$$

$$| h = y \cdot \sin \theta \Rightarrow$$

将 dF 对整个面积 A 积分, 合力 F ,

$$F = \int_A dF = \rho g \sin \theta \int_A y \cdot dA \quad (2.33)$$

面积矩, 静面积
(面积 A 对 x 轴的)

若 y_c —— 面积 A 的质心 C 的 y 坐标,

$$\text{则有: } \int_A y dA = y_c \cdot A$$

Eq. 2.33 is transform to,

$$F = \rho g y_c \cdot \sin \theta \cdot A = \rho g h_c A \quad (2.34)$$

上式表明: 静止液体作用在平板面上的压强

合力等于板面质心点上的压强与

上式表明：静止液体作用在平板面上的压强
合力等于板面质心点上的压强与
板面面积的乘积

2. 合力的作用点

由理论力学知识：

平行力系中各分力对某轴的力矩之和等于合力
对该轴的力矩。

假设合力作用点为 D，其坐标为 (x_D, y_D) 。

则由合力矩原理可得：

$$F \cdot x_D = \int_A x dF = \rho g \sin \theta \int_A x y dA$$

$$F \cdot y_D = \int_A y dF = \rho g \sin \theta \int_A y^2 dA$$

J_{xy} 为面积 A 对 x, y 两轴的惯性积；

J_x 为面积 A 对 x 轴的惯性矩；

$$\begin{aligned} \text{代入 (2.34) 式: } x_D &= \frac{J_{xy}}{A y_c} \\ y_D &= \frac{J_x}{A y_c} \end{aligned} \quad (2.35)$$

为了使用方便：可根据平行移轴原理。

$$\begin{cases} J_{xy} = J_{cxy} + x_c y_c A \\ J_x = J_{cx} + y_c^2 A \end{cases}$$

J_{cxy} 主惯性积 —— 是面积 A 对通过质心点 C
且平行于 x 轴、y 轴的两正交
轴线的惯性积。

J_{cx} 主惯性矩 —— 是对通过质心点且平行
于 x 轴的轴线的惯性矩。

代入 (2.35) 式：

$$x_D - x_c = \frac{J_{cxy}}{y_c \cdot A} \quad \text{若 } J_{cxy} = 0, \text{ (平板是对称)} \\ x_D = x_c$$

$$y_D - y_c = \frac{J_{cx}}{y_c \cdot A} \quad \downarrow J_{cx} > 0 \\ y_D > y_c$$

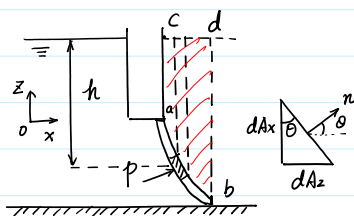
因此合力作用点的淹深总是大于质心点的淹深
 $h_D > h_c$

§2.7 Hydrostatic Forces on Curved Surfaces

由于液体作用在曲面上各点的压强方向都不相同。

所以在求合力时不能将微元面积 dA 上的压强

合力直接对面积积分, 而必须首先把 dA 上的力分解为水平方向和竖直方向的分量, 然后分别对同方向的分量积分。



1) 对于任意 z 维曲面 ab , 取微元面积 dA ,

液体作用在 dA 上的压强合力 dF ,

$$dF = \rho g h dA$$

将 dF 分解到 x, z 方向, 分别对面积 A 积分

得到压强合力在相应方向的分量

$$F_x = \int_A \rho g h \cos \theta \cdot dA \rightarrow dA_x$$

$$F_z = \int_A \rho g h \sin \theta \cdot dA \rightarrow dA_z$$

dA_x —— dA 在 x 方向的投影面积

dA_z —— dA 在 z 方向的投影面积

$$F_x = \rho g \int_{A_x} h dA_x = \rho g h_{xc} A_x$$

$$F_z = \rho g \int_{A_z} h dA_z = \rho g V$$

A_x —— 曲面 A 在 x 方向的投影面积

h_{xc} —— 面积 A_x 质心点的淹深。

V —— 曲面以上到液面所在水平面之间的体积, 也称
压力体
ooo

可见:

- ① 压强合力在 x 方向的投影分量 F_x 等于
液体作用在平面面积 A_x 上的压强合力。
- ② z 方向上的投影分量 F_z 等于压力体体积
中的液体重量。

Thus. the resultant force will be

$$F = \sqrt{F_x^2 + F_z^2}$$

the angle α between the action line
and horizontal line

$$\tan \alpha = \frac{F_z}{F_x}$$

F_x 的作用线通过水平分力在 A_x 上的作用力。

F_z 的作用线通过压力体 V 的质心

F 的作用线应通过以上两力线的交点,

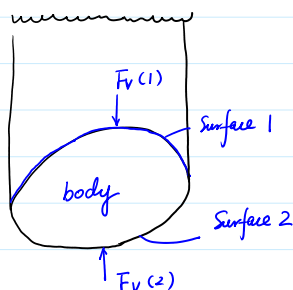
压强合力 F 的作用线与曲面的交点为
它的作用点。

* 作用在三维曲面上的压强合力可分解成
3个分量, F_x, F_y, F_z , 此时有 两个水平分量
 F_x, F_y , 求解方法与二维曲面 F_x 相同.

§ 2.8 Buoyancy and stability

1. Archimedes' Law of buoyancy 3rd Century B.C.

- ① A body immersed in a fluid experiences a vertical buoyant force equal to the weight of the fluid it displaces.
- ② A floating body displaces its own weight in the fluid in which it floats.

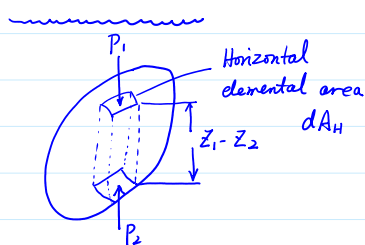


The body experiences a net upward force

$$F_B = \underbrace{F_v(2)}_{\text{(fluid weight over 2)}} - \underbrace{F_v(1)}_{\text{(over 1)}}$$

= weight of fluid equivalent to V (body volume)

Another Case,



$$\begin{aligned} F_B &= \int_{\text{body}} (p_2 - p_1) dA_H \\ &= \int_{\text{body}} (\rho g z_2 - \rho g z_1) dA_H \\ &= -\rho g \int_{\text{body}} h dA_H \\ &= \rho g \int_{\text{body}} h dA_H \\ &= \rho g V \end{aligned}$$

↳ (body volume)

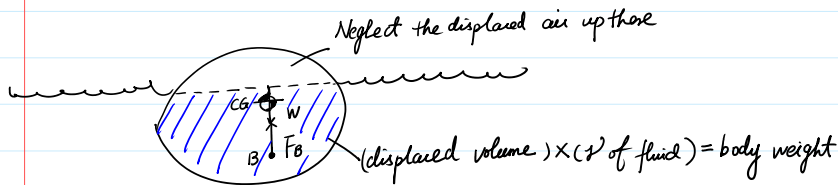
上述说明: 基于均匀比重的液体假设。浮种力
的作用线穿过压力体排开体积的
中心; 或者说: 只要液体具有均匀的

高度, 就可以求其质心。

通常用B或CB标记。

* 漂浮是特殊的例子。

仅有一部分体积浸没液体中, 剩余部分浮在自由液面上。



$$F_B = \rho \cdot \text{displaced volume} (g V) \\ = \text{floating-body weight}$$

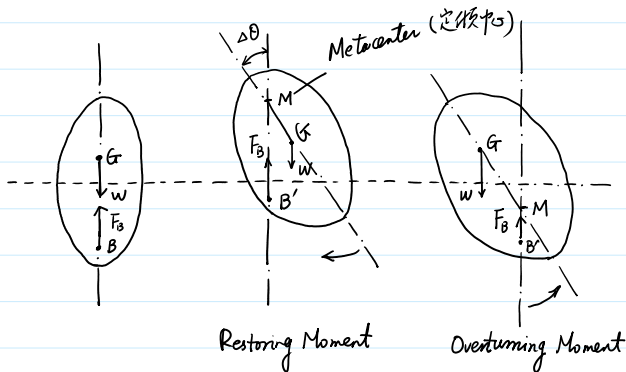
2. stability

如果一个物体的比重等于液体的比重相等

那么它处于“中间悬浮”状态。

根据阿基米德定理:

静止流体作用于物体的浮力, 大小等于物体排开的相同体积流体的重量, 方向铅垂向上, 且作用线通过浮心。(the center of buoyancy)



- ① The body's center of Mass G (重心)
the center of buoyancy B (浮心)

- ② 一般物体自身的重心与浮心并不重合
G and B are not always superposed.

- ③ A vertical line drawn upward from B' intersects the line of symmetry at a point M, called the metacenter.

- ④ If point M is above G,

$\overline{MG}$ 为“+” $\Rightarrow$ restoring moment

metacentric height 恢复力矩,

这个力矩可使浮体恢复到初始平衡位置。

此时它是稳定的。

If M is below G

$\overline{MG}$ 为 “-” $\Rightarrow$ Overturning moment

此时浮体是不稳定的，一个翻倾力矩

将纹浮体翻转。

Therefore, stability increases with increasing $\overline{MG}$

and $\overline{MG}$ value indicates the stability of the body.

§ 2.9 Pressure Distribution in Rigid-Body Motion

1. 压力分布的通用方程:

$$\nabla p = f(g-a) + \mu \cancel{\sigma^2 V}$$

刚体是所有粒子一起平动或转动,而无粒子

间的相对运动,也没有应力变形,

$$\nabla p = f(\vec{g} - \vec{a})$$

压力梯度的作用方向指向 $\vec{g} - \vec{a}$

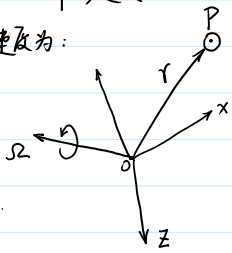
如果物体的质心为 O , 此点的平移速度

为 V_0 ，那么物体上任意一点 P 的速度为：

$$\vec{V} = \vec{V}_0 + \Omega \times r_0$$

Ω — 角速度矢量

r_0 — P 点的坐标位置.



对上式微分:

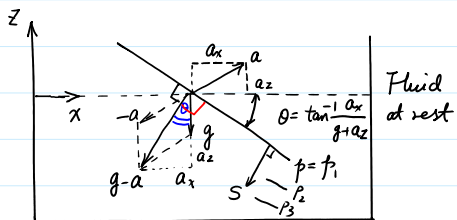
$$\vec{a} = \frac{d\vec{v}_0}{dt} + \underbrace{\omega \times (\omega \times \vec{r}_0)}_{\text{centripetal a.}} + \underbrace{\frac{d\omega}{dt} \times \vec{r}_0}_{\text{linear a.}}$$

* 上述3项很少同时应用于某一流体。

实际上流体很少做刚性运动, 除非长时间

束缚在有限容器内。

2. Uniform Linear Acceleration



$$\theta = \tan^{-1} \frac{a_x}{g + a_z}$$

$$\frac{dp}{ds} = \rho G, \quad G = \sqrt{a_x^2 + (g + a_z)^2}$$

上述压力分布与容器的尺寸、形状无关，只要液体连续地充满整个容器。

3. Rigid-Body Rotation

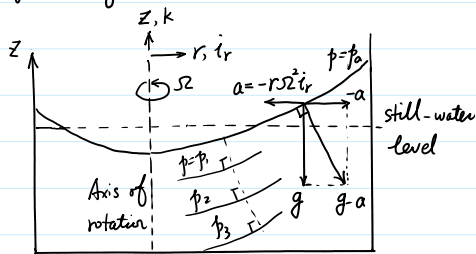


Fig. 2.22

Assumption ① only rotation with z axis, with no translation.

② the container rotate long enough at constant Ω so as to consider the fluid have reached to R-B rotation.

Thus, $\vec{\Omega} = \Omega \vec{k}$, $\vec{r}_0 = \vec{e}_r \cdot r$
 $\vec{\Omega} \times (\vec{\Omega} \times \vec{r}_0) = -r\Omega^2 \vec{e}_r$

For, rigid body rotation,

$$\nabla p = \rho(\vec{g} - \vec{a}) = \rho(-g\vec{k} + r\Omega^2 \vec{e}_r)$$

$$= \vec{e}_r \frac{\partial p}{\partial r} + \vec{k} \frac{\partial p}{\partial z}$$

$$\begin{cases} \frac{\partial p}{\partial r} = \rho r \Omega^2 \\ \frac{\partial p}{\partial z} = -\rho g \end{cases}$$

holding z constant, then

$$p = \frac{1}{2} \rho r^2 \Omega^2 + f(z)$$

$$\frac{\partial p}{\partial z} = 0 + f'(z) = -\rho g$$

$$f(z) = -\rho g z + C$$

thus, $p = \frac{1}{2} \rho r^2 \Omega^2 - \rho g z + C$

$p = p_0$ at $(r, z) = (0, 0)$, then $C = p_0$

The final desired distribution is

$$p = \frac{1}{2} \rho r^2 \Omega^2 - \rho g z + p_0$$

如果画出一个等压面，则 $p = p_1$

$$z = \frac{p_0 - p_1}{\rho g} + \frac{r^2 \Omega^2}{2g} = a + br^2$$